

Deepanwita Banerjee, PhD

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SUMMARY

Interdisciplinary R&D Scientist with a PhD in Systems Biology and extensive experience in bioinformatics, molecular biology, and multi-omics analysis. Proven track record in developing bioinformatics pipelines and integrating Jupyter-based workflows to accelerate the DBTL cycle. Strong foundation in mathematical modeling, biochemical reaction kinetics, and mammalian cell culture processes. Expert in genome-scale metabolic modeling and high-throughput experimental validation. Proficient in MATLAB and Python. Eager to contribute to Pfizer's team by leveraging expertise in mathematical modeling and process optimization.

EXPERIENCE

*Lawrence Berkeley National Laboratory (LBNL), **Research Scientist (Comp Bio), Mukhopadhyay Group***

Aug 2021 - Present

Bioinformatics Pipeline Development: Designed and implemented automated bioinformatics pipelines utilizing Jupyter Notebooks to process and analyze large-scale multi-omics datasets, reducing data turnaround time and increasing reproducibility.

Collaborative Development: Leveraged GitHub for version control and collaborative script development, ensuring a standardized codebase for analyzing microbial strain physiology and production metrics. **Integrated Omics Analysis:** Combined NGS-derived genomic data with physiological readouts to evaluate microbial strain performance under industrially relevant conditions, guiding the design of next-generation engineered strains. **Model-Driven Engineering:** Translated complex hypotheses from genome-scale metabolic models into experimental molecular biology designs, resulting in the construction of highly optimized microbial strains for isoprenol and indigoidine production. **Data Automation:** Integrated Jupyter-based analysis tools into automated workflows to streamline the transition from raw sequencing/metabolomics data to actionable biological insights.

*Joint Bioenergy Institute (JBEI) – Berkeley Lab, **Postdoctoral Fellow, Mukhopadhyay Group***

Oct 2018 - Aug 2021

Systems Biology Framework: Developed a scalable, model-integrated workflow to predict and validate strain engineering strategies for biofuels and bioproducts in *Pseudomonas putida* KT2440. **Predictive Modeling:** Successfully predicted growth-coupled production strategies for indigoidine; validated these predictions through extensive molecular biology and bioreactor cultivation. **NGS Application:** Used genomic characterization tools to verify the stability and efficiency of engineered pathways at scale.

EDUCATION

CSIR-National Chemical Laboratory (NCL), Pune, India

*PhD • **Systems Biology***

Research Focus: Developed a systems-level understanding of cell metabolism and enzymopathies. **Key Project:** Applied constraint-based modeling and molecular biology to study *Chromobacterium violaceum*, identifying key metabolic bottlenecks and proposing NAD⁺/NADH balance strategies to overcome antibiotic resistance. **Core Expertise:** Integrated computational predictions with wet-lab validation to map metabolic flux and identify novel targets for strain optimization.

SKILLS

Metabolic Modeling: COBRA Toolbox, MATLAB, COBRApy, Genome-scale metabolic models, Flux Balance Analysis, Minimal Cut-sets and Elementary Mode Analysis • **Data Analytics:** Jupyter Notebooks, Big data analysis • **Metabolic Engineering:** Computational strain optimization methods, growth-production tradeoff evaluation • **Bioprocess Physiology:** Strain physiology in industrially relevant cultivation regimes

HONORS & AWARDS

US Patent Application 18/976,085

USPTO • Jun 2025

Genetically modified pseudomonas host cells and methods useful for producing isoprenol. (Co-inventor)

US Patent 11,767,521

USPTO • Sep 2023

Co-inventor - Genetically modified bacterial cells and methods useful for producing indigoidine.

JBEI annual award in Outreach and Community contribution.

Joint BioEnergy Institute • 2020

SPOT recognition award by Berkeley Lab.

Joint BioEnergy Institute • Nov 2020

JBEI annual award for Research contribution

Joint BioEnergy Institute • Nov 2021